
Partitioned Surrogate Consensus for Multi-Channel Convolutional Dictionary Updates

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Abstract

The expensive step in convolutional dictionary learning is not convolution itself, but the rank carried by the training sources in each Fourier-domain normal equation. Existing updates handle that rank through equality-split ADMM, consensus ADMM, conjugate gradients, iterated Sherman–Morrison, or accelerated majorization, and are usually derived separately. We express these choices with one weighted source-cover template: a local model is either an exact quadratic resolvent or a Lipschitz majorizer, while the cover, metric application, coordinates, and memory specify an update. The same state machine reproduces four independent SPORCO implementations on real images; the maximum trajectory and final-dictionary deviations are at most 2.25×10^{-16} and 4.54×10^{-16} , respectively. Choosing the rank source also expresses a multi-channel dictionary with shared maps and a shared dictionary with channel-specific maps; both match references and end at 0.1697 and 0.2474 of their zero-code functional. The framework further gives a crossed channel–image cover that preserves the objective exactly and places rank- C and rank- K resolvents in one consensus step. It reaches 0.2508 of its initial functional and collapses to reference consensus within 5.97×10^{-15} . Finally, a joint channel–image block DFT exposes a multi-channel volume/consensus equivalence. A controlled sweep identifies the resulting wall-time versus progress tradeoff and the limits of the new factorization.

1 Introduction

Convolutional dictionary learning (CDL) represents training signals by

$$s_k \approx \sum_{m=1}^M d_m * x_{m,k}, \quad \min_{D,X} \frac{1}{2} \sum_k \left\| \sum_m d_m * x_{m,k} - s_k \right\|_2^2 + \lambda \sum_{m,k} \|x_{m,k}\|_1, \quad (1)$$

with each filter restricted to a prescribed support and unit norm. Alternating optimization makes the coefficient step a familiar convolutional sparse coding problem, but leaves a harder dictionary step. At a spatial frequency, its normal operator is an identity plus a sum of K outer products. Thus the data term has rank at most K , whereas the corresponding sparse-coding system has the rank-one structure that admits a single Sherman–Morrison update [6, 12].

The standard remedies look unrelated. Equality-split ADMM solves the complete rank- K system by iterated Sherman–Morrison (ISM) or conjugate gradients (CG); consensus ADMM gives each image a rank-one local dictionary; and FISTA replaces the solve by an accelerated projected-gradient step [1, 2, 6]. The apparent distinction is partly representational: a block-circulant “3D” formulation becomes consensus after a DFT over the image index [6]. This suggests treating source factorization, quadratic approximation, and coordinates as choices inside a common update, rather than as names of separate algorithms.

Channels make that common structure more useful. A color signal can use multi-channel filters and one map per atom, or a shared single-channel filter with a map for every channel. These models have different rank sources: images in the first case and channel–image pairs in the second. Both were derived previously [5, 11]; carrying existing solvers to color is therefore a baseline, not a contribution.

This paper makes four claims.

- A weighted source-cover formulation and one exact/MM update grammar generate equality ADMM (ISM or CG), singleton consensus, and FISTA. An authored state machine reproduces all four independent reference trajectories, without calling the reference dictionary solvers.
- The rank-source definition expresses both principal multi-channel representations without changing the update. It also makes the image/channel rank duality explicit.
- For a shared dictionary, an overlapping *crossed cover* uses image-wise channel groups and channel-wise image groups. Half weighting counts every fidelity term exactly once. We are not aware of a prior CDL realization of this particular cover; the claim is deliberately narrower than novelty of consensus or overlapping covers in general.
- A block DFT over both channels and images converts a joint block-circulant volume problem to replicated consensus. The derivation extends the known image-volume identity and predicts the same constraint scaling observed numerically.

Our scope is the dictionary block with the sparse-code step held fixed. The empirical study uses deterministic reference-parity tests, one real-image color instance, and a small controlled regime sweep; Section 7 states the resulting limits on generalization.

2 Rank sources in grayscale and multi-channel CDL

Fix the coefficient maps and vectorize the feasible dictionary as d . For a set of *rank sources* $\mathcal{Q}$, the dictionary subproblem is

$$\min_d F(d) + \iota_{C_{PN}}(d), \quad F(d) = \sum_{q \in \mathcal{Q}} f_q(d), \quad f_q(d) = \frac{1}{2} \|A_q d - s_q\|_2^2, \quad (2)$$

where C_{PN} is the support-plus-unit-norm set and $P_{C_{PN}}$ restricts support then normalizes each atom. At frequency ν ,

$$\nabla^2 \hat{F}_\nu = \sum_{q \in \mathcal{Q}} \hat{x}_{q,\nu} \hat{x}_{q,\nu}^H. \quad (3)$$

The update is therefore controlled by $Q = |\mathcal{Q}|$, not by the visual dimensionality of a signal.

For grayscale training, $q = k$ and $Q = K$. With a C -channel atom $d_m = (d_{1,m}, \dots, d_{C,m})$ and a map $x_{m,k}$ shared by channels, the source is still $q = k$: the same rank- K factor is applied to all C dictionary channels. With one shared single-channel atom d_m and maps $x_{c,m,k}$, the source is $q = (c, k)$ and $Q = CK$. The implementation obtains the latter case by flattening channel and image into a single source index, not by adding a color-specific solver. A joint-sparsity penalty across $\{x_{c,m,k}\}_c$ changes the coefficient step but leaves Eq. (2) in the same $q = (c, k)$ form; it is outside the fixed coefficient step studied here.

This bookkeeping also clarifies the familiar $K \leftrightarrow C$ duality: K supplies rank to a grayscale dictionary update, while C supplies rank to multi-channel sparse coding. In the shared-dictionary representation they appear together as the product CK , which motivates factorizations that do not flatten away their two meanings.

3 A partitioned surrogate-consensus template

3.1 Weighted covers and exact resolvents

Let $\mathcal{G} = \{G_1, \dots, G_J\}$ cover $\mathcal{Q}$. Associate a nonnegative scalar w_j with group G_j and require

$$\sum_{j:q \in G_j} w_j = 1 \quad \text{for every } q. \quad (4)$$

Define $F_j(d) = w_j \sum_{q \in G_j} f_q(d)$, or equivalently stack $\sqrt{w_j} A_q$ and $\sqrt{w_j} s_q$ into A_j, s_j .

Proposition 1 (Objective preservation). *Under Eq. (4), $\sum_j F_j(d) = F(d)$ for every d . Consequently,*

$$\min_{\{d_j\}, g} \sum_{j=1}^J F_j(d_j) + \iota_{\mathcal{C}_{PN}}(g) \quad \text{s.t.} \quad d_j = g \quad (5)$$

has exactly the same feasible objectives and minimizers as Eq. (2).

Scaled consensus ADMM applied to Eq. (5) gives one recurrence:

$$d_j^{t+1} = (A_j^H A_j + \rho I)^{-1} \{A_j^H s_j + \rho(g^t - u_j^t)\}, \quad (6)$$

$$v_j^{t+1} = \alpha d_j^{t+1} + (1 - \alpha)g^t, \quad (7)$$

$$g^{t+1} = P_{\mathcal{C}_{PN}} \left(\frac{1}{J} \sum_j (v_j^{t+1} + u_j^t) \right), \quad (8)$$

$$u_j^{t+1} = u_j^t + v_j^{t+1} - g^{t+1}. \quad (9)$$

At each frequency, Eq. (6) has metric

$$\rho I + \sum_{q \in G_j} w_j \hat{x}_q \hat{x}_q^H, \quad (10)$$

whose local rank $r_j = |G_j|$ determines the numerical kernel. Rank one uses a Sherman–Morrison solve; larger ranks use ISM or warm-started CG. The recurrence is unchanged.

A full cover ($J = 1$) is equality ADMM. Singleton groups are conventional consensus. Intermediate disjoint groups are data-block consensus. Thus partition size changes the same linear operator, rather than selecting a separately derived method. Direct or unitary block coordinates may be used around Eqs. (6)–(9); unitary coordinates preserve the objective and residual norms by Parseval’s identity.

3.2 Majorized local models

The same source sum also admits the scalar upper model

$$Q_L(d; y) = F(y) + \langle \nabla F(y), d - y \rangle + \frac{L}{2} \|d - y\|_2^2, \quad L \geq \|\nabla^2 F\|_2. \quad (11)$$

Table 1: Factorization grid generated by the template. V: validated against an independent implementation; K: published configuration; N: filled CDL configuration studied here; E: coordinate equivalence. A dash denotes an untested configuration, not an impossibility.

Local model / memory	Full source block	Singleton sources	Disjoint source groups	Crossed channel-image cover
Exact / ADMM dual	equality ADMM: ISM, CG (V)	rank-one consensus (V)	data-block consensus [3] (K)	rank- C and rank- K consensus (N)
Scalar MM / Nesterov	FISTA and BPG-M variants [4] (V/K)	APG consensus [10] (K)	–	–
Unitary coordinates	tilted/volume interpretations (K)	image-volume = consensus (E)	blockwise = trans-forms	joint (C, K) volume = consensus (E)

Minimizing this model plus the indicator gives

$$d^{t+1} = P_{C_{PN}}(y^t - L^{-1}\nabla F(y^t)), \quad (12)$$

$$\tau_{t+1} = \frac{1}{2}(1 + \sqrt{1 + 4\tau_t^2}), \quad y^{t+1} = d^{t+1} + \frac{\tau_t - 1}{\tau_{t+1}}(d^{t+1} - d^t). \quad (13)$$

With Nesterov memory, this is the FISTA dictionary update. Without memory it is projected gradient. Exact resolvents use replica dual memory; scalar majorizers use no dual. These are the two local-model choices in the implementation, and both consume the same source operator and the same $P_{C_{PN}}$.

Table 1 is a grammar, not a solver catalogue. Its independent choices are the representation (which defines q), source cover, local model, metric application, coordinates, memory, and placement of the constraint. The final choice is fixed here: $\iota_{C_{PN}}$ always remains in the shared projection.

4 Consequences of the common algebra

4.1 A crossed channel-image cover

For a shared dictionary and channel-specific maps, let $\mathcal{Q} = \{(c, k) : 1 \leq c \leq C, 1 \leq k \leq K\}$. Consider

$$G_k = \{(c, k) : c = 1, \dots, C\}, \quad H_c = \{(c, k) : k = 1, \dots, K\}, \quad (14)$$

with weight $1/2$ for each of the $K + C$ groups. Every source occurs in one G_k and one H_c , so Proposition 1 applies exactly. Local G_k solves have rank C and local H_c solves rank K . This redundant cover exposes both sides of the channel/image duality inside a single dictionary step.

Proposition 2 (Rank-work interpolation). *For the $O(r^2)$ source dependence of ISM, the per-frequency rank-work proxy is $(CK)^2$ for one full block, CK for singleton consensus, and*

$$KC^2 + CK^2 = CK(C + K) \quad (15)$$

for the crossed cover. Hence the crossed-to-full ratio is $1/C + 1/K$.

The full arithmetic also scales with the number of filters and spatial frequencies, and consensus adds replica storage and averaging. Equation (15) therefore predicts an interpolation, not a universal speedup. It also explains why the cover can become attractive as both C and K grow.

Arbitrary disjoint data blocks are already used in consensus CSC [3], and singleton majorized consensus has also been studied [10]. Our initial disjoint-group candidate was withdrawn after finding the former work and is retained only as a breadth experiment. A search of the CDL literature found no use of the overlapping two-way cover in Eq. (14); we make no claim that weighted overlapping consensus is new outside this CDL instantiation.

4.2 Joint channel-image volume is consensus

The image-volume equivalence extends to both source indices. Let $Q = CK, U = F_C \otimes F_K$ be the unitary two-dimensional block DFT, and construct the doubly block-circulant operator

$$B_{(r_c, r_k), (c, k)} = A_{(r_c - c \bmod C, r_k - k \bmod K)}. \quad (16)$$

Let $E = e_{(0,0)} \otimes I_m$ inject d into the first of Q dictionary blocks. With $T_n = U \otimes I_n$ and $T_m = U \otimes I_m$,

$$T_n B T_m^H = \text{diag}(\tilde{A}_1, \dots, \tilde{A}_Q), \quad (17)$$

$$T_m E d = Q^{-1/2} \mathbf{1}_Q \otimes d. \quad (18)$$

Therefore Parseval's identity transforms $\frac{1}{2} \|B E d - s\|_2^2$ into

$$\frac{1}{2} \sum_{\ell=1}^Q \left\| \tilde{A}_\ell \frac{d}{\sqrt{Q}} - \tilde{s}_\ell \right\|_2^2, \quad (19)$$

a separable fidelity with replicated dictionary blocks. Absorbing the common $\sqrt{Q}$ scale yields consensus. The image-only result is recovered at $C = 1$; using $F_C \otimes F_K$ shows that treating a multi-channel training set as a joint volume changes coordinates, not the optimization class. Full proofs of Propositions 1–2 and Eqs. (17)–(18) are in Appendix A.

For a random $C = 3, K = 2$ block construction, the logged relative objective error is 0, the off-diagonal block energy ratio after the transform is 3.69×10^{-16} , and the replicated-constraint error is 1.33×10^{-16} . These checks include the normalization in Eq. (18); they are not inferred from the plotted CDL runs.

5 Empirical study

5.1 Protocol and independent references

All images come from MIRFLICKR25k [7]. We use seeded image selection, center-square crops, area resizing, scaling to $[0, 1]$, and the SPORCO Tikhonov high-pass filter with regularization 5 and padding 8. Alternation follows the reference protocol: one ADMM CBPDN coefficient iteration and one dictionary iteration per outer step, $\lambda = 0.1$, fixed sparse-code penalty 5.5, dictionary penalty 1, unit relaxation, and no adaptive penalty. Every reported dictionary is projected onto $\mathcal{C}_{PN}$ before evaluating the functional.

The reference-fidelity instance uses three 32×32 grayscale images, $M = 8$ filters of support 5×5 , and 30 outer steps. CG uses relative tolerance 10^{-3} and cap 1000; FISTA uses $L = 42$, Nesterov memory, and no backtracking. The color instance uses four real 64×64 RGB images, $C = 3$, $M = 64$, 8×8 filters, and 120 outer steps. A fresh SPORCO ConvBPDNDictLearn run at $\lambda = 0.1$ is made for each representation on the identical input and initialization. The authored template imports SPORCO's indexing, FFT, projection, low-rank algebra, and fixed CBPDN stage, but does not call its ISM, CG, consensus, or PGM dictionary-solver classes; those classes run only in the reference harness [13].

5.2 Reference fidelity and multi-channel convergence

Figure 1 overlays every authored instance with a separately executed SPORCO solver. The ISM, CG, and FISTA trajectories and dictionaries agree to stored floating-point equality; singleton consensus has maximum trajectory deviation 2.25×10^{-16} and dictionary error 4.54×10^{-16} . No sign or permutation alignment is applied. Final filter norms lie within 2.2×10^{-16} of one. Recovering all four families through the same update engine is important: ISM and CG change only the metric application in Eq. (6), while consensus changes the cover and FISTA changes the local model and memory.

Both color representations start from the same zero-code functional $F_0 = 398.357322$ but define different models, so their endpoints should not be compared as if they optimized the same variables. The multi-channel dictionary/shared-map model reaches $F_{120} = 67.604607$ ($0.1697F_0$); the shared-dictionary/channel-map model reaches $F_{120} = 98.571973$ ($0.2474F_0$). Both are below the registered $0.6F_0$ threshold and match their own reference trajectory and dictionary exactly. The relative change between the means of steps 101–110 and 111–120 is 0.581% and 0.419%, respectively. Figure 2 shows that the match holds throughout alternation, not only at the endpoint.

Table 2: Primary results. Deviation is maximum pointwise functional-trajectory deviation / final relative dictionary error. Gray rows use $T = 30$; color rows use $T = 120$. The dagger marks a degenerate-cover comparison with SPORCO consensus; all other color deviations compare the displayed full instance directly with its matched reference.

Method	Representation	Setting: local model; cover; metric / memory	δ_F	δ_D	F_T
Equality ADMM-ISM	gray	exact; one K block; ISM / dual	0	0	7.733985
Equality ADMM-CG	gray	exact; one K block; CG / dual	0	0	7.734265
Consensus ADMM	gray	exact; K singletons; rank-one SM / dual	2.25×10^{-16}	4.54×10^{-16}	7.805753
FISTA	gray	scalar MM; full source sum; gradient / Nesterov	0	0	8.017120
Full exact	C -channel shared X	D , exact; one K block; ISM / dual	0	0	67.604607
Full exact	shared D , channel X	exact; one CK block; ISM / dual	0	0	98.571973
Data-block consensus	C -channel shared X	D , exact; image pairs; ISM / dual	$6.10 \times 10^{-16}^\dagger$	$1.09 \times 10^{-14}^\dagger$	68.202614
Crossed cover	shared D , channel X	exact; K rank- C + C rank- K groups; ISM / dual	$4.28 \times 10^{-16}^\dagger$	$5.97 \times 10^{-15}^\dagger$	99.906734

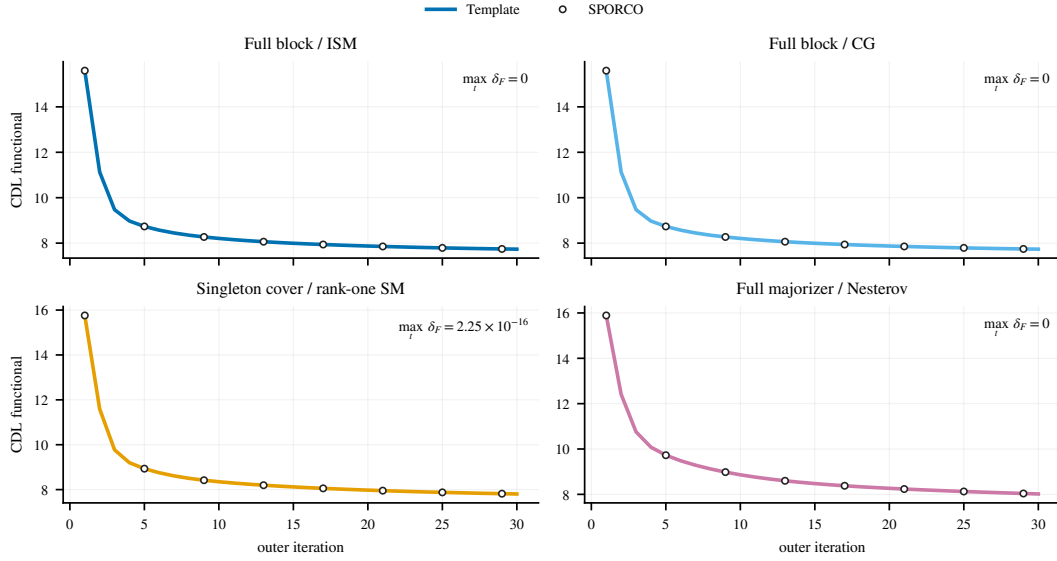


Figure 1: Reference fidelity on real grayscale images. Markers are independently executed SPORCO trajectories; lines are instances of the common template. The maximum pointwise relative deviation is annotated in each panel.

5.3 Crossed-cover behavior and collapse

The crossed cover has seven groups with ranks $(3, 3, 3, 3, 4, 4, 4)$. The logged weight sum is exactly one for all 12 sources. It reaches $F_{120} = 99.906734 = 0.2508F_0$, with a late-window relative change of 0.555% and all filter norms equal to one within 4.4×10^{-16} . Replacing Eq. (14) by singleton channel-image groups yields SPORCO consensus with trajectory deviation 4.28×10^{-16} and dictionary error 5.97×10^{-15} . This degeneration tests the cover machinery, weighting path, and multi-channel reshape against an external endpoint.

Figure 3 also prevents a stronger but unsupported performance claim. At a fixed 120 steps, full exact updates reach the lowest functional, singleton consensus the highest, and the crossed cover lies between them. Its contribution is a factorization with two semantically meaningful local views, not a new best endpoint.

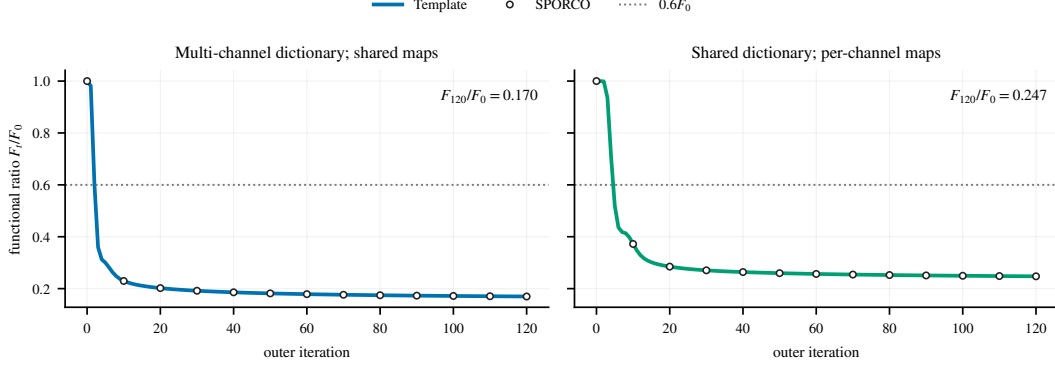


Figure 2: Multi-channel instances at $C = 3, K = 4, M = 64$. Each model is normalized by its common zero-code functional. Open markers are matched SPORCO runs at the same $\lambda = 0.1$.

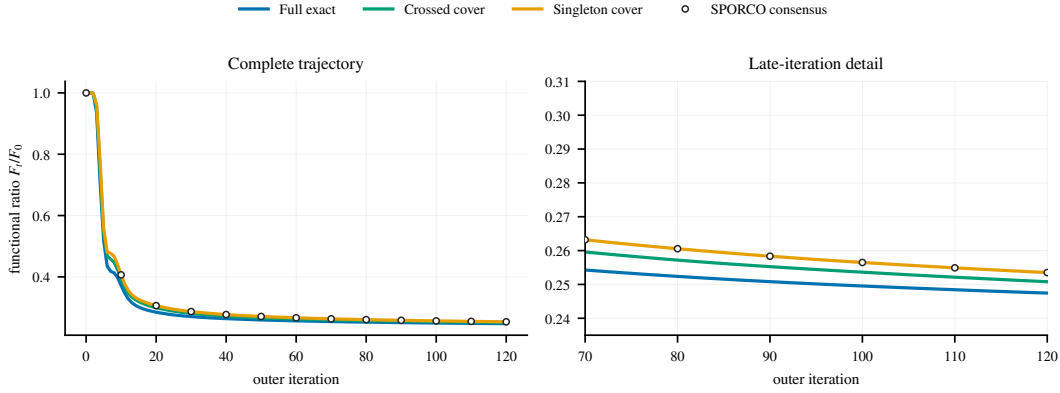


Figure 3: The crossed cover on the shared-dictionary/channel-map model. Singleton markers coincide with the independent SPORCO consensus run. The right panel resolves the late-iteration ordering.

5.4 Which factorization suits which regime?

We vary $C \in \{2, 4\}$, $K \in \{2, 4, 8\}$, and nominal channel correlation in $\{0, 0.9\}$, giving 12 regimes and 48 complete template runs. The data are energy-matched mixtures of real high-pass MIRFLICKR images; empirical mean correlations span -0.038 – 0.006 and 0.805 – 0.858 in the two conditions. Each run uses $M = 16$, 5×5 filters, 32×32 signals, and 30 outer steps. For FISTA, $L = 14CK$.

The selection map in Figure 4 separates early wall-clock progress from progress per outer step. FISTA has the lowest estimated time to $0.6F_0$ in 10 of 12 regimes; the full exact solve wins the two small, highly correlated cases. Conversely, the full exact solve has the lowest F_{30} in all 12 regimes. At the largest tested source grid, $C = 4, K = 8$, the crossed cover reduces time per outer step by 31.0% at low correlation and 20.9% at high correlation, while its F_{30} is only 0.70% and 0.70% above full exact, respectively. Its rank-work proxy is 384 rather than 1024, consistent with Proposition 2.

The practical rule is consequently conditional. FISTA is attractive when early wall-clock reduction matters and a conservative scalar majorizer is acceptable. A full exact resolvent gives more decrease per alternating step when CK is moderate. Singleton consensus minimizes local rank and exposes maximum parallelism at the cost of CK replicas. The crossed cover is an intermediate option for shared dictionaries when preserving image-wise and channel-wise local subproblems is valuable; its arithmetic advantage appears only once both dimensions are nontrivial.

5.5 Compute cost

All experiments ran on one 16-vCPU Linux host with 128 GB RAM, single-thread BLAS/FFT settings, and no GPU. The 66 logged solver runs sum to 380.4 seconds of wall time; this includes

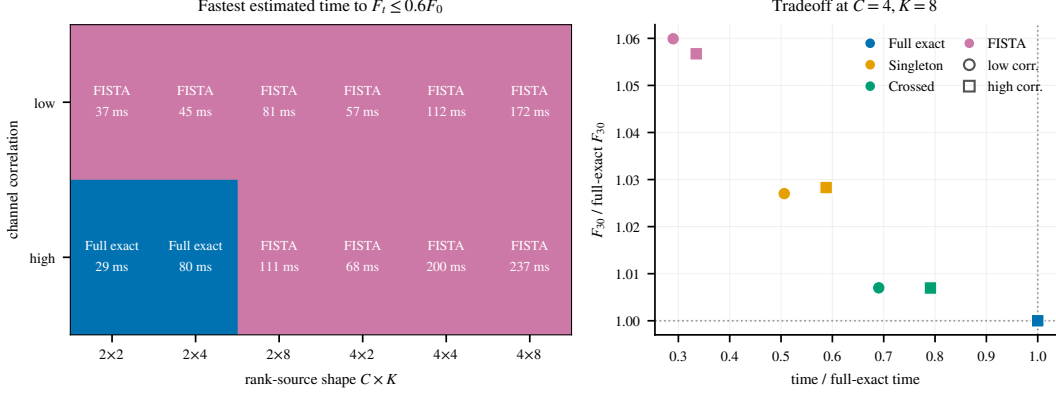


Figure 4: Left: fastest estimated wall time to $0.6F_0$; each cell reports the method and time. Right: time/endpoint tradeoff at the largest tested C, K , normalized by full exact within each correlation condition. Timings are single-process measurements, not hardware-independent benchmarks.

all references, collapse checks, the withdrawn grouped configuration, and the complete sweep. The large color candidate times range from 19.2 seconds for the full multi-channel-dictionary instance to 49.413 seconds for the redundant crossed cover. These are implementation measurements rather than claims about distributed execution, for which replica communication would be decisive.

6 Relation to prior methods

The Fourier-domain CDL algorithms unified here were systematically compared by Garcia-Cardona and Wohlberg [6]; the same work proved the image-volume/consensus identity and derived multi-channel updates. Multi-channel CDL was developed further by Garcia-Cardona and Wohlberg [5], while Wohlberg [11] distinguished shared and multi-channel color representations. Our baseline color results deliberately reproduce these known models. The contribution is the source-cover/MM abstraction, its crossed instantiation, and the joint (C, K) coordinate identity.

Alternative factorizations occupy nearby parts of the grid. Choudhury et al. [3] split large training data into disjoint blocks; Silva and Rodriguez [10] combined image consensus with accelerated proximal gradients; and Chun and Fessler [4] developed convergent block proximal-gradient majorizers and update schedules. Local spatial training [9] and online first/second-order CDL [8] address locality or streaming rather than the fixed batch dictionary subproblem. Our template does not subsume their sampling, boundary, or online-memory mechanisms.

7 Limitations, failure audit, and broader impact

Optimization scope. The propositions establish objective preservation and coordinate equivalence, not global convergence of alternating CDL. The support-plus-sphere projection and joint problem are nonconvex. We match the one-step coupling of the reference implementation; different inner schedules, adaptive penalties, relaxation, masks, or boundary operators may change the empirical ordering. The joint-sparsity color model is recognized by its $q = (c, k)$ dictionary update, but its distinguishing coefficient penalty is not implemented.

Empirical scope. Reference parity is deterministic numerical verification, so repeated seeds would not add a meaningful confidence interval to that claim. The color anchor uses one four-image selection, and the regime study uses one initialization and constructed correlations rather than natural hyperspectral data. Timing has no repetitions or uncertainty bars. The selection rule should therefore be read as a mechanism check, not a population-level performance ranking. The crossed cover never beats the full solve in F_T at a fixed iteration count in our sweep.

Failure audit. The first candidate beyond the standard endpoints was a size-two disjoint image partition. A literature audit found that data-block consensus was already instantiated by Choudhury

et al. [3]; we withdrew novelty, retained its $F_{120} = 68.202614$ result, and registered the crossed cover before running it. A first CG harness execution also failed while serializing an iteration-statistic field; the full recovery matrix was rerun after fixing the field lookup. Neither event changed numerical thresholds or inputs. The literature-based originality claim remains scoped to the specific two-way CDL cover.

Broader impact. A common update grammar can reduce duplicated implementation effort and help choose lower-compute solvers, improving access to sparse imaging methods. Like generic image representations, learned dictionaries could enter beneficial scientific imaging or undesirable surveillance pipelines. This work releases no trained model and does not expand data collection; its most direct effect is numerical efficiency. Energy savings from better factorization are positive but small at the experimental scale here.

8 Conclusion

The rank in a CDL dictionary update can be treated as a collection of sources rather than as a fixed obstacle. Covering those sources, choosing an exact or majorized local quadratic, applying the resulting metric, and selecting coordinates turns four named algorithms and two channel representations into instances of one update grammar. The crossed channel–image cover and the joint block-DFT equivalence are consequences of that grammar. Exact reference parity and objective-preserving collapse tests anchor the abstraction; the regime sweep shows why no single factorization dominates both wall time and per-step progress.

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A Proofs

A.1 Proof of Proposition 1

Using finite sums and Eq. (4),

$$\sum_{j=1}^J F_j(d) = \sum_{j=1}^J \sum_{q \in G_j} w_j f_q(d) = \sum_{q \in \mathcal{Q}} \left(\sum_{j: q \in G_j} w_j \right) f_q(d) = \sum_{q \in \mathcal{Q}} f_q(d) = F(d). \quad (20)$$

On the feasible set of Eq. (5), $d_j = g$ for all j , so its objective equals $F(g) + \iota_{\mathcal{C}_{PN}}(g)$. Conversely, every feasible d in Eq. (2) lifts to $d_j = g = d$. The feasible objectives and minimizers therefore correspond in both directions.

A.2 Derivation of the common exact update

The scaled augmented Lagrangian of Eq. (5), up to constants, is

$$\sum_j \left[\frac{1}{2} \|A_j d_j - s_j\|_2^2 + \frac{\rho}{2} \|d_j - g + u_j\|_2^2 \right] + \iota_{\mathcal{C}_{PN}}(g). \quad (21)$$

Differentiating each unconstrained d_j quadratic gives Eq. (6). The g minimizer is $\text{prox}_{\iota_{\mathcal{C}_{PN}}/(J\rho)}(J^{-1} \sum_j (v_j + u_j))$; an indicator is invariant to positive scaling, yielding Eq. (8). The scaled dual ascent is Eq. (9). Thus full, singleton, disjoint, and crossed covers change only (A_j, s_j) and replica count.

A.3 Proof of Proposition 2

SPORCO’s ISM recurrence applies r_j successive rank-one corrections and forms interactions with all earlier corrections, giving quadratic source-rank dependence $\sum_j r_j^2$. Full flattening has one $r_1 = CK$ block. Singleton consensus has CK blocks of rank one. The crossed cover has K groups of rank C and C groups of rank K , so $\sum_j r_j^2 = KC^2 + CK^2$. Division by $(CK)^2$ gives $1/K + 1/C$.

A.4 Proof of the joint block-DFT equivalence

Index the $Q = CK$ blocks by the finite product group $\mathbb{Z}_C \times \mathbb{Z}_K$. Equation (16) is convolution on this group with matrix-valued kernel $\{A_{c,k}\}$. Its irreducible characters are the rows of $U = F_C \otimes F_K$, so the convolution theorem gives

$$(U \otimes I_n) B (U^H \otimes I_m) = \text{diag}(\tilde{A}_1, \dots, \tilde{A}_Q), \quad (22)$$

proving Eq. (17). Since a unitary DFT maps the impulse $e_{(0,0)}$ to $Q^{-1/2} \mathbf{1}_Q$,

$$(U \otimes I_m)(e_{(0,0)} \otimes I_m) d = Q^{-1/2} \mathbf{1}_Q \otimes d, \quad (23)$$

which proves Eq. (18). Left multiplication by the unitary T_n preserves the residual norm. The transformed fidelity is therefore Eq. (19), and all transformed dictionary blocks are equal after the common scaling. This is precisely the consensus constraint.

B Implementation and reproducibility details

Single state machine. The implementation stores a source cover, one projected dictionary g , and one local/dual pair per exact group. For every outer iteration it: (i) executes one fixed SPORCO CBPDN step; (ii) reshapes coefficient maps into the representation’s source indexing; (iii) applies Eq. (6) to every group using rank-one SM, ISM, or CG, followed by Eqs. (8)–(9); or applies Eqs. (12)–(13) for a majorizer; and (iv) passes the projected dictionary back to CBPDN. There is no call from this state machine to a SPORCO dictionary-update class.

Commands. From the workspace root, the logged experiments are reproduced with

```
PYTHONPATH=code/sporco:proposal/code .venv/bin/python \
  proposal/code/experiments.py recovery
PYTHONPATH=code/sporco:proposal/code .venv/bin/python \
  proposal/code/experiments.py color
PYTHONPATH=code/sporco:proposal/code .venv/bin/python \
  proposal/code/experiments.py crossed
PYTHONPATH=code/sporco:proposal/code .venv/bin/python \
  proposal/code/experiments.py equivalence
PYTHONPATH=code/sporco:proposal/code .venv/bin/python \
  proposal/code/experiments.py regime
.venv/bin/python proposal/code/make_artifacts.py
cd proposal
pdflatex -interaction=nonstopmode -halt-on-error report.tex
bibtex report
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
cd ..
.venv/bin/python proposal/code/final_audit.py
```

The all experiment option runs the five experiment groups in the same order. JSON files contain configurations, trajectories, functional components, timings, norms, hashes, and deviations; NPZ files contain full dictionaries and trajectories.

Data selections. The grayscale anchor uses MIRFLICKR image identifiers (23628, 15194, 15938) with seed 20260728. The color anchor uses (17764, 14383, 9731, 21072) with seed 20260828. The regime pool uses seed 20261028 and records all 40 identifiers in logs/regime_sweep.json. Initial dictionaries use subsequent, separately logged seeds. No test image is used to tune an experimental threshold.

Software and assets. Runs use Python 3.14.4, NumPy 2.5.1, and the supplied SPORCO 0.2.3 development tree. SPORCO is BSD-3-Clause licensed. MIRFLICKR makes images available for research under the individual licenses stored with its metadata; we do not redistribute a modified collection. The official NeurIPS 2025 style was obtained from the conference style archive because the expected local style file was absent. Source hashes and versions are stored in every experiment record or the artifact manifest.

C Complete traceability and audit

The timestamped framework record is code/framework_plan.{md,json}; its executable reader is code/framework_plan.py. The initial timestamp is 2026-07-28T13:14:38Z. The literature-triggered crossed-cover revision is timestamped 2026-07-28T13:25:14Z, before the crossed run. The disjoint-group and CG serialization events are preserved in attempts_log.md.

Table 3 maps every quantitative part of the paper to its primary record. The generated logs/results_derived.json contains source hashes and every aggregate used in prose; logs/artifact_manifest.json hashes the derived record, result macros, and figures. The post-compilation logs/final_audit.json records threshold checks, JSON/NPZ equality, finite-array checks, manifest verification, build diagnostics, and the final PDF hash.

Table 3: Primary quantitative records.

Content	Record under proposal/logs/
Four grayscale references	recovery_*.json/.npz
Two color representations	color_multichannel_dictionary.*, color_shared_*
Disjoint grouped breadth	color_grouped_consensus_*
Crossed cover and collapse	color_crossed_*
Coordinate equivalence	multichannel_equivalence.*
Regime characterization	regime_sweep.*
Paper aggregates and hashes	results_derived.json, artifact_manifest.json
Final submission audit	final_audit.json

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2. **Limitations.** Does the paper discuss the limitations of the work performed by the authors? **Answer:** [Yes] . **Justification:** Section 7 discusses optimization, empirical, timing, representation, and originality limitations.
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5. **Open access to data and code.** Does the paper provide open access to the data and code, with sufficient instructions to reproduce the main results? **Answer:** [Yes] . **Justification:** The submission artifact includes authored code, reference harness, logs, dictionaries, and commands. MIRFLICKR is publicly available under its per-image research licenses.
6. **Experimental setting/details.** Does the paper specify all training and test details needed to understand the results? **Answer:** [Yes] . **Justification:** Section 5 specifies all anchor and sweep settings; exact image identifiers and environment versions are in Appendix B.
7. **Experiment statistical significance.** Does the paper report error bars or other appropriate information about statistical significance? **Answer:** [No] . **Justification:** Parity is a deterministic equality check. The characterization sweep uses one initialization per cell and is presented as a mechanism study; Section 7 explicitly limits the inference and reports no confidence intervals.
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12. **Licenses for existing assets.** Are existing code and data assets properly credited and are their licenses respected? **Answer:** [Yes] . **Justification:** Appendix B identifies SPORCO’s BSD-3-Clause license and MIRFLICKR’s per-image license terms, and cites both assets.
13. **New assets.** Are new assets introduced in the paper well documented? **Answer:** [Yes] . **Justification:** The new implementation, experiment harness, machine-readable records, hashes, commands, and failure audit are included and mapped in Appendix C.
14. **Crowdsourcing and research with human subjects.** Does the paper include the required participant information? **Answer:** [NA] . **Justification:** There is no crowdsourcing or human-subject research.
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